



A Systematic Comparison of Large and Small Language Models in *Standalone, Hybrid & Multi-Agent* Architectures

Doğukan Topçu · Abdulbaki Çeltik · Berkhan Kargılı · Meriç Kelemeden

Advisors: Dr. Hüseyin Ünlü · Prof. Dr. Onur Demirörs

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1 Abstract / Summary

Measurement-first platform benchmarking **ten architectures in three families** across **five public benchmarks** under a unified observability contract (accuracy · latency · cost · energy · carbon). We introduce **EATS** (Efficiency-Accuracy Trade-off Score) as an architecture-agnostic metric for comparing quality and efficiency together. Across the study, **selective routing** gives the strongest overall trade-off, while novel **Blackboard-style** and **swarm** designs show task-specific strengths on reasoning benchmarks.

2 Background / Literature Review

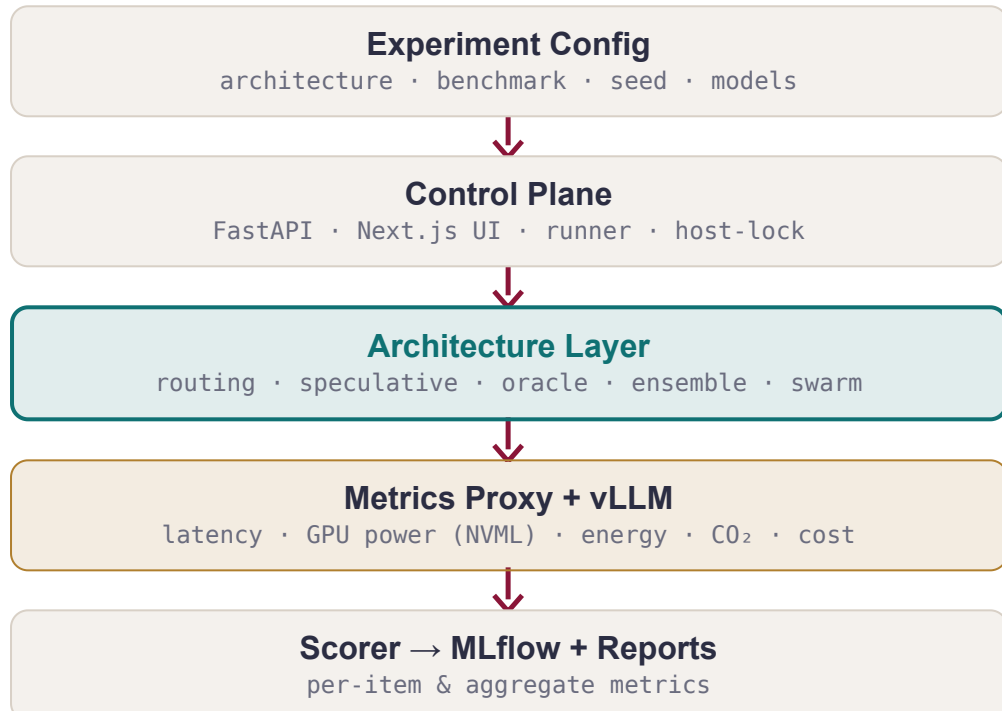
LLMs deliver state-of-the-art reasoning but impose high latency, cost, and energy. **SLMs** are cheaper and data-sovereign, yet trail on broad reasoning tasks.

A systematic literature review of **54 collaborative SLM–LLM studies** indicates that, while hybrid and multi-agent designs are becoming more common, **cost and energy are still rarely treated as primary evaluation outcomes** alongside accuracy.

RQ1	Performance. How do collaborative SLM/LLM systems compare with standalone LLMs on reasoning & classification?
RQ2	Efficiency. How do hybrid & multi-agent designs affect cost, latency, and energy?
RQ3	Unified metric. Does EATS give a stable, architecture-agnostic ranking of the quality–efficiency trade-off?
RQ4	Empirical grounding. Do measured comparisons corroborate the trends from our 54-study SLR?

3 Measurement-First Platform

Every architecture consumes the same Query and returns the same Response carrying **accuracy, latency, tokens, cost, energy & CO₂** per query. The architecture layer is the *only* interchangeable component.



TIER	HOST	VRAM	MODELS
SLM	NVIDIA L4	24 GB	Gemma/Qwen/LLaMA 3–4B
Mid LLM	NVIDIA RTX PRO6000	96 GB	20–35B dense + MoE
Heavy LLM	NVIDIA H200	141 GB	LLaMA-70B, gpt-oss-120B

4 EATS · Efficiency–Accuracy Trade-off Score

A single score comparing any architecture against the standalone-LLM baseline.

$$\text{EATS} = \frac{\text{Accuracy}}{\text{Accuracy} + \text{Penalty}}$$

$$\text{Penalty} = 0.5 \cdot \text{Cost} + 0.3 \cdot \text{Latency} + 0.2 \cdot \text{Energy}$$

each ÷ standalone-LLM baseline → LLM = 1

0.5 is the theoretical ceiling for the standalone LLM.

5 Energy, Carbon & Cost

Per-request measurements via GPU power sampling (NVIDIA NVML) at the inference host.

$$\text{Cost} = \text{GPU rate} \cdot \text{time}$$

$$\text{Latency} = \Delta t \text{ per request}$$

$$E = P_{\text{GPU}} \cdot \Delta t \text{ kWh}$$

$$\text{CO}_2 = k \cdot E$$

$$k = 380 \text{ gCO}_2/\text{kWh (EU avg)} \cdot E \text{ in J} \div 3.6 \times 10^6$$

6 Benchmarks & Protocol

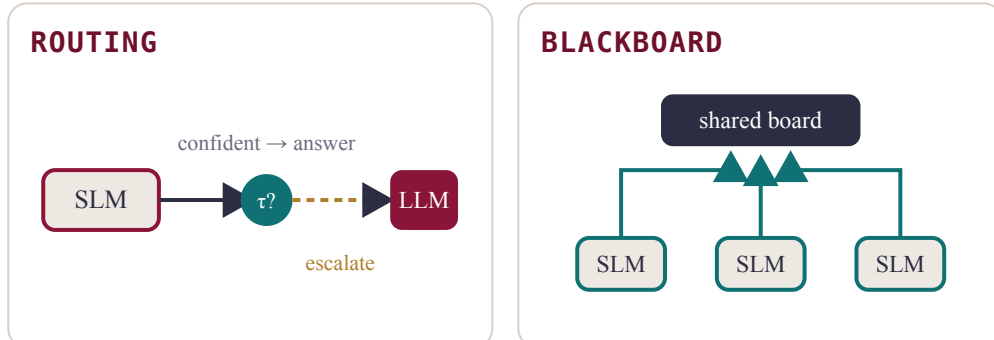
Five benchmarks span the **capability dimensions most cited in our companion SLR**, letting us test whether efficiency gains persist as tasks shift from knowledge retrieval to chain-of-thought math and adversarial fact-checking.

BENCHMARK	WHAT IT MEASURES
MMLU	Broad knowledge — 57 academic disciplines
ARC-C	Scientific reasoning — multi-step physical-world science
GSM8K	Mathematical reasoning — grade-school math, free-form
HellaSwag	Commonsense inference — sentence completion
TruthfulQA	Adversarial truthfulness — calibration vs. persuasive falsehoods

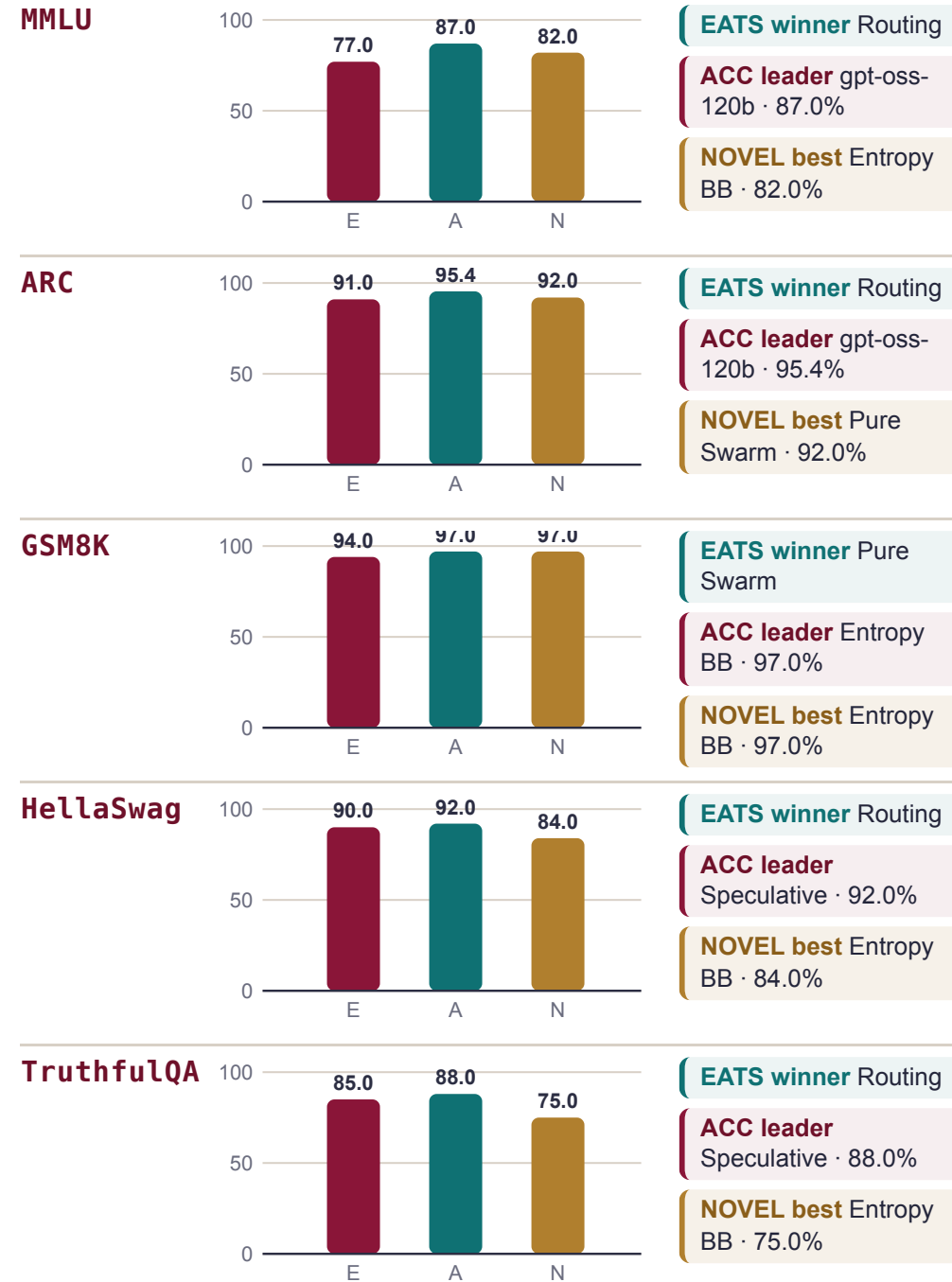
7 Architecture Taxonomy

Ten architectures in three families. † marks novel project contributions introduced in the current study.

FAMILY	ARCHITECTURE	TYPE	KEY MECHANISM
STAND	Single-Model	Baseline	direct inference
STAND	MoE	Baseline	internal sparse expert activation
HYBRID	Routing	Literature	confidence SLM→LLM escalation
HYBRID	Speculative	Literature	SLM drafts; LLM rewrites suffix
HYBRID	Debate (POA)	Literature	proponent–opponent–arbitrator
HYBRID	Active Oracle †	Novel	targeted LLM sub-queries
HYBRID	Blackboard †	Novel	bid-based task board; LLM sweeper fallback
HYBRID	Entropy Blackboard †	Novel	uncertainty-penalised bidding
MULTI	Ensemble	Literature	majority or confidence-weighted voting
MULTI	Pure Swarm †	Novel	SLM-only peer coordination; no LLM online



8 Benchmark Readouts



9 EATS Landscape

Canonical EATS for the latest qualified run of each architecture; the standalone row shows the best MoE baseline on each benchmark.

ARCHITECTURE	MMLU	ARC	GSM8K	HSWAG	TQA
Routing	0.82	0.88	0.58	0.85	0.81
Speculative	0.81	0.85	0.51	0.83	0.78
Standalone (MoE)	0.46	0.49	0.49	0.48	0.46
Blackboard †	0.54	0.64	0.64	0.49	0.34
Entropy BB †	0.52	0.62	0.65	0.48	0.38
Active O. †	0.57	0.83	0.56	0.61	0.46
Debate POA	0.34	0.43	0.41	0.31	0.29
Pure Swarm †	0.53	0.66	0.68	0.50	0.48
Ensemble	0.22	0.29	0.26	0.24	0.20

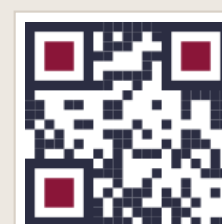
Darker = higher EATS · bold = benchmark winner · tinted rows / † = novel contribution.

10 Conclusions

- RQ1** Collaborative systems are competitive with standalone LLMs; the strongest results come from routing, speculative decoding, and novel coordination on reasoning-heavy tasks.
- RQ2** Hybrid escalation gives the clearest cost and energy gains, while multi-agent coordination usually pays a latency penalty.
- RQ3** EATS gives a consistent cross-architecture ranking: routing leads four benchmarks, and Pure Swarm leads GSM8K.
- RQ4** The measurements broadly corroborate the SLR: SLM-first escalation is the strongest general pattern, while coordination and entropy-aware bidding help on task-specific cases.

SELECTED REFERENCES

[1] Hao et al. **Hybrid SLM–LLM Inference**, 2024 · [2] Oh & Kim, **Uncertainty-Aware Hybrid Inference**, 2025 · [3] Shao et al. **Division-of-Thoughts**, 2025 · [4] Kwon et al. **vLLM: PagedAttention**, SOSP 2023 · [5] Companion SLR (54 studies, PRISMA 2020) · [6] MMLU, ARC, GSM8K, HellaSwag, TruthfulQA.



Code & experiments
github.com/DogukanTopcu